

# LINUX Training Contents 6 and 7 version

## History:

- History of UNIX
- History of LINUX
- Different types of operating systems
- Flavors of open source OS
- Difference b/w RHEL, Centos and Fedora, Debian, Ubuntu etc....

## Introduction of Linux:

- Architecture of Unix/Linux
- Advantages of Linux OS
- Application of Linux OS
- Installation methods
- File system hierarchy structure
- Type of consoles
- Types of desktops

## History of commands:

- Basic commands
- Filter commands
- Different editors
- Man pages

## Linux File System:

- Types of files
- Hard link & Soft link
- File system types
- Difference between file systems
- Mount techniques
- Characteristics of file system

## Partition management:

- Need of Partitioning
- Partition techniques
- Partition creation tools
- Normal partition creation
- Swap partition creation
- Hexa code conversions
- Partition encryption techniques
- Partition troubleshooting

## LVM:

- PV create, VG create, LV create
- LV extend/reduce
- Migration
- LV snapshots
- Backup and restore LVM data file

- More about LVM

## User & Group administration:

- Important files related to user and group
- User administration
- Group administration
- Password parameters
- Password aging policies

## File permissions:

### Basic file permissions

- Read
- Write
- Execute

### Advanced file permissions

- SUID
- SGID
- STICKY BIT

### Access Control List

## Disk Quotas:

- Enabling Quotas on partitions
- Quota database creation
- Quotas based on blocks and inodes

## RAID:

- Software RAID
- Hardware RAID
- RAID levels (0,1,4,5)

## Backup and recovery:

- Backup types
- Backup tools
- SCP & RSYNC

## Remote administration:

- Telnet
- SSH
- Vnc viewer

## Security of linux:

- SELinux
- Iptables

## Boot recovery and kernel parameters:

- Grub recovery
- Root password recovery
- Kernel parameters

### Package management:

- Redhat Package Management (RPM)
- Yellow Dog Updater Modifies (YUM)

### Process management:

- Interactive process
- Daemon process
- Automate process

### Basic Networking:

- Types of layers
- Difference between protocols
- IP address classes
- Assign static IP
- Client DNS trouble shooting

### Network File Systems (NFS) services:

- Overview of NFS
- Information required to configure NFS server
- Configuration of NFS server
- Configuration of NFS client

### File Transfer Protocol (FTP) services:

- Overview of FTP
- Files/packages required to configure NFS server
- Configuration of FTP server
- Configuration of FTP client

### SAMBA services:

- Overview of SAMBA server
- Packages required to configure SAMBA server
- Configuration of samba server
- Configuration of samba client

### Web service with Apache

- Webserver with HTML Basic Programming

### DHCP services:

- Overview of DHCP server
- Information required to configure DHCP server

- Configuration of DHCP server
- Configuration of DHCP client

### DNS services

- Overview of DNS server
- Packages required to configure DNS server
- Configuration of DNS server
- Configuration of DNS client

### Web service with Apache-Tomcat:

- Overview of Apache – Tomcat
- Packages required to configure Tomcat server
- Configuration of Tomcat server
- Import the war file

### Network Information Service (NIS):

- Overview of NIS
- Difference between NIS
- Information required configuring NIS server
- Configuration of NIS server
- Configuration of NIS client

### Mail server (send mail):

- Overview of POP and IMAP
- Information required to configure sendmail server
- Configuration of sendmail server
- Configuration of control panel for client

### Data base server (SQL):

- Overview of different kind of databases
- Information required to configure SQL server
- Configuration of SQL server
- Configuration of SQL client

### Syslog server:

- Overview of syslog server
- Configuration of syslog server
- Configuration of syslog client
- Log rotation

### Basic Shell scripting

- Example Programs with Real Time

**Additional Free Topics JAVA and PHO Developers Deployment Applications**

Ex: eclipse , Navicat , Ireports , MYSQL,APACHE TOAMCAT, Real Time Scenarios